

EDUCATION

Duke University
M.S. in Statistical Science

Durham, NC
2019–2021

Peking University
B.S. in Statistics

Beijing
2014–2018

RESEARCH INTERESTS

My recent research interests are the scientific foundations of Large Language Models. I am interested in developing better evaluation, interpretability, and theory for this emerging technology. In particular, I study Transformers from a statistical perspective, using tools like causal inference and high dimensional statistics.

PUBLICATION & PREPRINTS

- [Jiajun Song](#), Zhuoyan Xu, and Yiqiao Zhong, [Out-of-distribution generalization via composition: a lens through induction heads in transformers](#), *Proceedings of the National Academy of Sciences*, January 2025
- Chi Zhang, [Jiajun Song](#), Siyu Li, Yitao Liang, Yuxi Ma, Wei Wang, Yixin Zhu, and Song-Chun Zhu, [Proposing and solving olympiad geometry with guided tree search](#), *Arxiv*, December 2024
- [Jiajun Song](#) and Yiqiao Zhong, [Uncovering hidden geometry in transformers via disentangling position and context](#), *Arxiv*, October 2023

PROFESSIONAL EXPERIENCE

Beijing Institute for General Artificial Intelligence
Researcher

Beijing, China
Aug 2022 – Present

- Investigated LLMs through symbolic reasoning tasks and induction head analysis, revealing the composition mechanism that enables OOD generalization.
- Conducted research on AI for Maths, including automated generation and solution of IMO-level geometry problems.
- Developed an interpretable decomposition of Transformer embeddings, uncovering low-dimensional and structured components that enhance understanding of model representations.

Warner Music Group
Data Scientist

New York, NY, USA
Jun 2021 – Aug 2022

- Developed monitoring systems for music streaming data using Markov-Switching Autoregressive Models and Bayesian online changepoint detection.
- Performed causal inference analyses on large-scale panel data to evaluate the impact of playlist exposure on streaming performance.

Etsy
Machine Learning Engineer Intern

New York, NY
May 2020 – Aug 2020

ACADEMIC PROJECTS

- [Analyzing the Effect of Early Adversities in the Life Expectancy of Wild Baboons](#) 2021
Survival Analysis, Causal Inference Advisor: Fan Li
- [Assessing the Racial Disparities in NYC Policing](#) 2021
Multilevel Poisson Regression, Spatial Statistics Advisor: David B. Dunson
- [Investigating Mortality Rates from Cardiovascular Pediatric Surgery in STS Public Reporting](#) 2020
Bayesian Joint Modeling, Bayesian Calibration
- [Python Package: Stochastic Gradient Hamiltonian Monte Carlo](#) 2020
Hamiltonian Monte Carlo, Package Development Advisor: Cliburn Chan

TEACHING

- **Teaching Assistant** at Duke University Spring 2021
Introduction to Mathematical Statistics (STA661), Scott C. Schmidler
- **Teaching Assistant** at Duke University Fall 2020
Theory of Statistical Inference (STA532), Janson Xu
- **Teaching Assistant** at Duke University Fall 2019
Introduction to Mathematical Statistics (STA661), Meimei Liu

SKILLS

- **Programming:** C, Python, R, Stan, SQL, PyTorch, JAX, React
- **Developer Tools:** Git, Docker, Kubernetes, Slurm, Cloud Computing
- **Language:** Mandarin (Native), English (Fluent)

RELEVANT COURSES

- **Undergraduate level:** Linear Algebra & Geometry (A+), Calculus (A+), Multivariate Calculus (A+), Theory of Group, Rings and Fields (A+), Algebra (A), Statistical Learning (A), Probability (A+),
- **Graduate level:** Causal Inference (A+), Real Analysis (A-), Measure Theory (A-), Statistical Inference (A), Deep Learning (A+), Statistical Programming & Computing (A), Bayesian Methods and Modern Statistics (A), , Programming, Data Structures, and Algorithms in C++ (A)

AWARDS

- Selected in Applied Mathematics Program for Elite Students, Peking University 2015–2018
- 1st prize in National High School Mathematics Olympic 2014